UiT The Arctic University of Norway

Technical requirement specification

Virtual Planetarium

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1 Introduction

The Virtual Planetarium (VP) pilot project was originally intended to run on a high-end computer with a high-resolution VR headset, but two years after the project start and after testing of the developed application, it seems that the partners will be able to run the VP on a "low-budged" system, containing of consumer VR headsets and computers, with reduced quality. This would be beneficial as the partners can perform testing and introduction of the technology in their region with the SMEs. What this "low-budged" system contains of and how-to set it up, is described in 2 Hardware.

2 Hardware

2.1 VR Headset

The Virtual Planetarium is designed for PCVR and is running in SteamVR.

Meta Quest 3 is standalone device with onboard CPU and GPU; however, the integrated hardware will not be powerful enough to run the VP. The Quest 3 is compatible for PCVR using the Meta Quest Link application and a high-speed USB-C 3.2 cable, and will then be able to run SteamVR applications as other headsets.

Please see the PC requirements to use *Meta Quest Lin*k (NOTE: This is NOT the PC requirements for running the VP). Set up Meta Quest Link.

Table 1 - Meta Quest 3

	Name	Description	Price	Website
Required	Meta Quest 3	512GB	€569.00	Meta Store
Required	Meta Quest Link cable	USB-C 3.2, 5 meters, for connection to PC	€104.99	Meta Store
Additional	Meta Quest Elite Strap	Flexible and adjustable head strap	€84.95	Meta Store
Additional	Meta Quest Elite Strap	With battery for extended user time	€154,99	Meta Store
Additional	Meta Quest 3 facial interface	Silicone	€49.99	Meta Store

2.2 VR Computer

It is recommended to use a desktop computer; however a laptop can be used but the performance will be limited and the system will generate a lot of heat. The laptops also have mobile versions of the GPUs that does not have the same performance as the full-size GPUs. If laptop is to be used, it needs to be with either an RTX 4090 or RTX 5090 GPU.

RTX 4090: Laptops from ASUS Laptops from GIGABYTE

RTX 5090 laptops are to be released in March 2025.

Examples on desktops from:

DELL HP ASUS CORSAIR MSI

OS: Win 10 / Win 11

Processor: Intel i7, or Higher / AMD Ryzen 7, or Higher*

Graphics: Nvidia RTX 3090, or Higher (minimum 20 GB VRAM)*

Memory: 32 GB DDR4 RAM, or Higher

Connection: USB-C 3.2 port, or better

Storage: 60 GB available space

Network: Broadband Internet connection

DirectX: Version 12

Sound Card: Any

VR Support: SteamVR, OpenXR, Seated, Standing, Roomscale. Valve Index, HTC Vives and Pico. Oculus headsets with Oculus Runtime, Meta headsets with Meta Quest Link.

Additional Notes:

Requires a 64-bit processor and operating system.

Virtual Desktop not recommended (Questlink/Airlink Fully Supported)

*CPU and GPU options are listed in Table 2.

Table 2 - CPU and GPU

	Minimum	**OK experience**	**Best experience**
GPU NVIDIA	RTX 2080, RTX 2080S, RTX 2080Ti, RTX 3060, RTX 3060Ti, RTX 4060, RTX 4060Ti	RTX 3070Ti, RTX 3080, RTX 3080Ti, RTX 3090, RTX 3090Ti, RTX 4070, RTX 4070S	RTX 4070Ti, RTX 4080, RTX 4080 S, RTX 4090, RTX 5080, RTX 5090
GPU AMD	RX 5700 XT, RX 6700 XT	RX 6800, RX 6800 XT, RX 6900 XT, RX 695 0XT	RX 7900 XT, RX 7900 XTX RX 9000 series
CPU INTEL	intel i7 - 10x00 Series or above intel i9 - 10x00 Series or above	Intel i7 - 12x00 Series or above intel i9 - 12x00 Series or above	intel i7 - 13x00 Series or Above intel i9 - 13x00 Series or above
CPU AMD	AMD Ryzen 7 AMD Ryzen 9	AMD Ryzen 7 AMD Ryzen 9	AMD Ryzen 7 AMD Ryzen 9